

Module 01: Systems in Robotics

Learning Objectives

1. Define **Systems** within the context of Robotics.
2. Analyze how **Systems** interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of **Systems**.

Introduction

This module explores **Systems**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. **Systems** is a critical component of the 8-part Active Inference spine, bridging the gap between Planning and Agents.

Key Concepts

1. Systems as a Markov Blanket Boundary

How does **Systems** define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in **Systems** must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of **Systems** drive the perception-action loop?

Applications

In Robotics, we see **Systems** manifest in: * **Specific Example 1:** A quadrotor flight control system represents the complete system as a state-space model with 12 states (position,

velocity, orientation, angular rates), where the system's Markov blanket is defined by four rotor speed commands (active states) and IMU plus barometer readings (sensory states); the A , B , C , D matrices of the linear state-space representation encode the generative model that the onboard Extended Kalman Filter uses to predict and correct state estimates at 400 Hz, making the system boundary mathematically explicit. * **Specific Example 2:** An industrial robotic workcell comprising a KUKA iiwa manipulator, a conveyor belt, and a vision system is modeled as a coupled dynamical system where each subsystem has its own state-space representation; the control-estimation framework treats the full workcell as a single generative model with block-diagonal dynamics (each subsystem evolves independently) and off-diagonal coupling terms (the conveyor speed affects the timing of the robot's pick motion), enabling coordinated free energy minimization across the entire system.

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.